JK Lakshmipat University Jaipur
INSTITUTE OF ENGINEERING AND TECHNOLOGY
Mid Term Examination, September 2026
B. Tech. (Section H), Semester I, 2026-27

Roll No...... 579 .....

Course code: AS1108 Course title: Applied physics Time: 90 mins Max. Marks: 20

Instructions to students:
1. Do not write anything other than your roll number on question paper.
2. Read the all the questions carefully and made necessary assumption.
3. Non-programmable scientific calculator is allowed.

Q. 1 a. Sketch the spherical coordinate system, clearly identifying the three unit vectors and their directions. (CO1, BTL 1, 2.0)

b. For planar polar coordinates $(r, \emptyset)$, write the unit vectors $\hat{r}$ and $\hat{\emptyset}$ in Cartesian-form and determine their time derivatives. (CO1, BTL 2, 2.0)

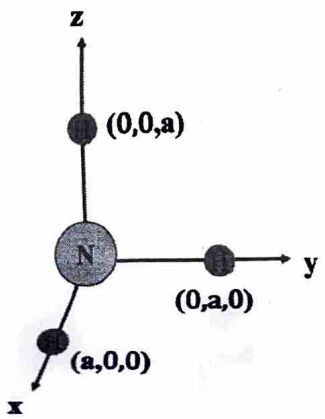
Q. 2 During a flood-relief operation, a drone is flying horizontally at constant speed and carries an emergency medical package. The drone releases the package when it is 800 m horizontally from a rescue platform. The platform is 20 m above the ground, and the package reaches the platform 6.0 s after release. Neglect air resistance and take g=9.8 m/s^2.

a. Determine the horizontal speed and the height of the drone above the ground when the package is released. (CO1, BTL 3, 2.0)

b. Determine the magnitude and direction of the velocity of the package just before it reaches the target. (CO1, BTL 3, 2.0)

OR

An ammonia (NH_3) molecule is modelled as a system of four point masses. The nitrogen atom (mass 14 units) is at the origin with the three hydrogen atoms (each mass 1 unit) located on the three axis as shown in the figure below.



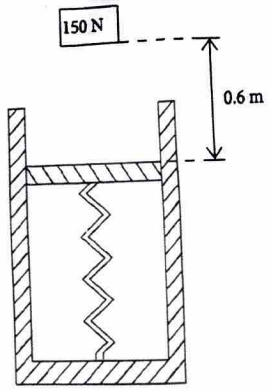
a. What is the position of the centre of mass of the molecule? (CO1, BTL 2, 2.0)
b. This molecules undergoes a chemical reaction and an H^+ ion (mass 1 unit) now attaches to the molecule. Determine the coordinates (x, y, z) of the location where the H^+ ion must attach so that the centre of mass of the resulting ammonium ion ($[NH_4]^+$) is at the origin. (CO1, BTL 4, 2.0)

Q. 3 The position vectors of a point in two different frames **S** and **S′** are $(4t+3)\hat{\imath} + (4t^2+t+1)\hat{\jmath} + (5t-3)\hat{k}$ and $(3t+3)\hat{\imath} + (4t^2+1)\hat{\jmath} + (4t-3)\hat{k}$. If both the frames are inertial in nature, then

a. Find the velocity of **S′** frame of reference. (CO1, BTL 3, 2.0)
b. Prove that the acceleration is the same in both the reference frames. (CO1, BTL 3, 1.0)
c. Suppose at some instant the particle is at rest in **S** frame. Would it also be at rest in **S′**? Explain. (CO1, BTL 5, 1.0)

Q. 4 Consider a cylinder-piston arrangement where a spring is placed as shown in the figure below. A block of weight 150 N is allowed to fall on piston from height 0.6 m If the stiffness of spring is 1.20 N/mm, determine:

a. the compression of spring by using the principle of conservation of energy. (CO2, BTL 2, 3.0)
b. the maximum force exerted by the spring on the piston. (CO2, BTL 3, 1.0)



Q. 5 Three particles each of masses 10 g, 20 g, and 40 g are initially moving with the speeds 20 cm/s along X-, Y-, and Z-axes, respectively. The particles collide simultaneously. After the collision, the 40 g particle comes to rest, while the 20 g particle moves with a velocity $(5\hat{\jmath} + 5\hat{k})$ *cm*/s. Determine the velocity of the 10 g particle after collision. Does the collision change the velocity of the centre of mass of the three-particle system? Justify your answer. (CO2, BTL 4, 4.

All the best!